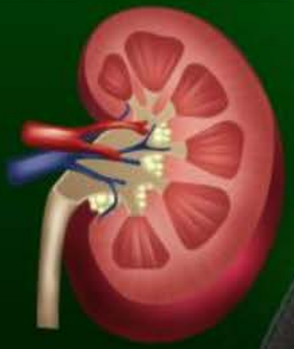
VICTORY 2027

The Nervous System



LECTURE BIOLOGY

Upasana Ma'am



Topics To Be Covered

1. BRAIN & its parts
2.

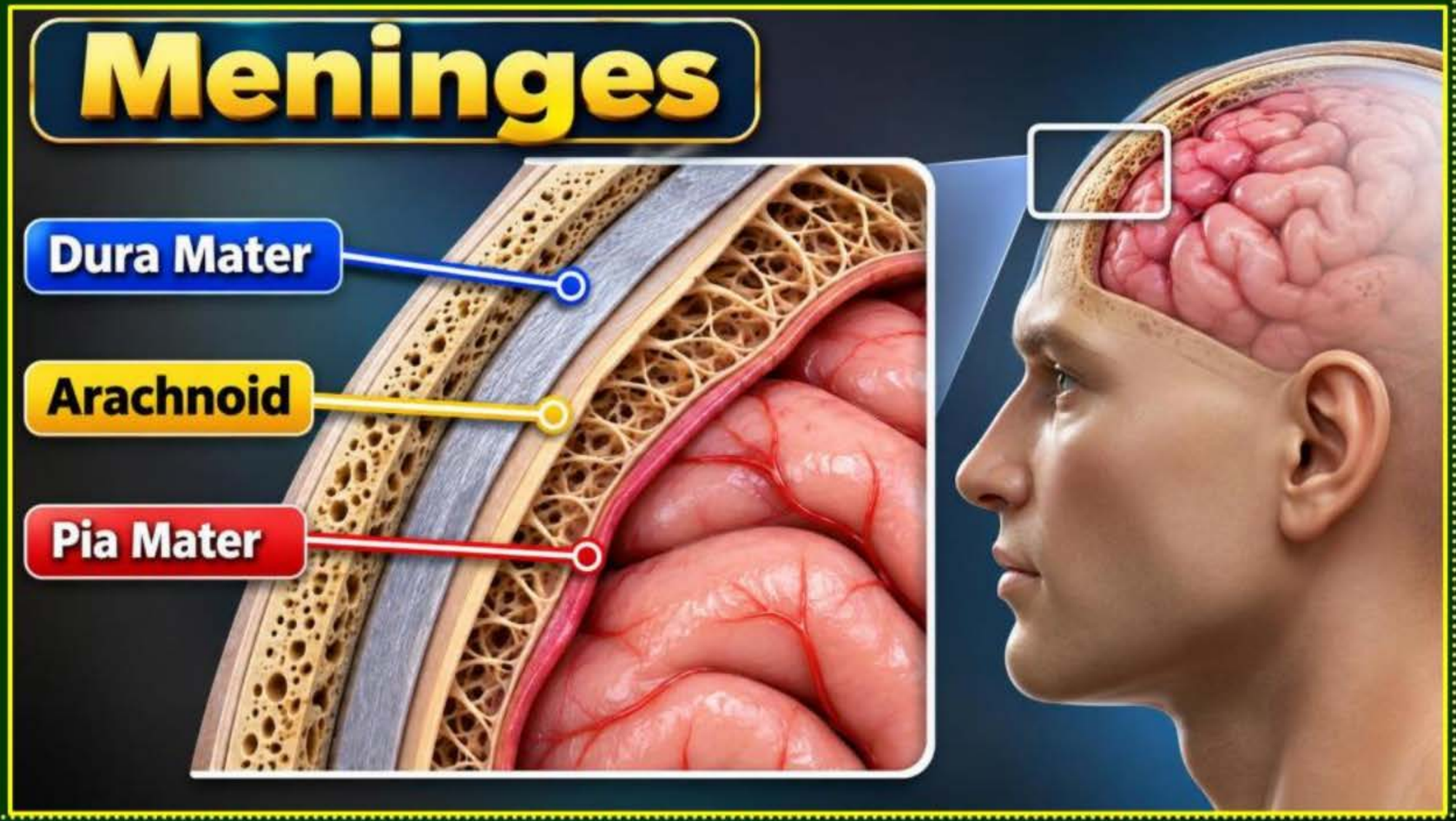
5-7 min

imp

Baatcheet

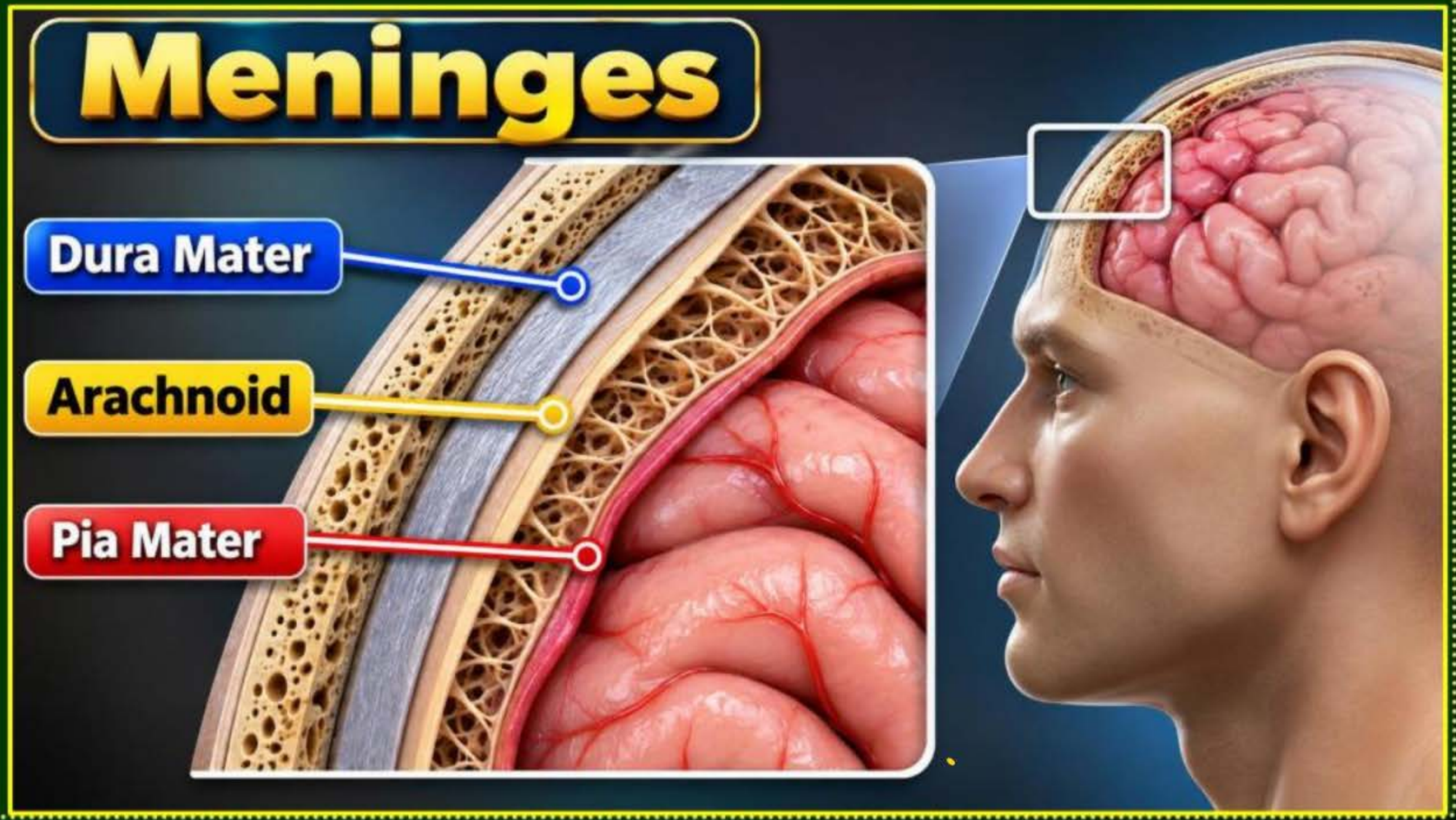
THE BRAIN

(i) Dura mater - the outermost tough fibrous membrane (dura : tough, mater: mother).



THE BRAIN

(ii) Arachnoid - the thin delicate middle layer giving a web-like cushion (arachne: spider).



THE BRAIN

(iii) Pia mater - the innermost highly vascular membrane, richly supplied with blood (pia : tender).

Meningitis is the inflammation of the meninges.

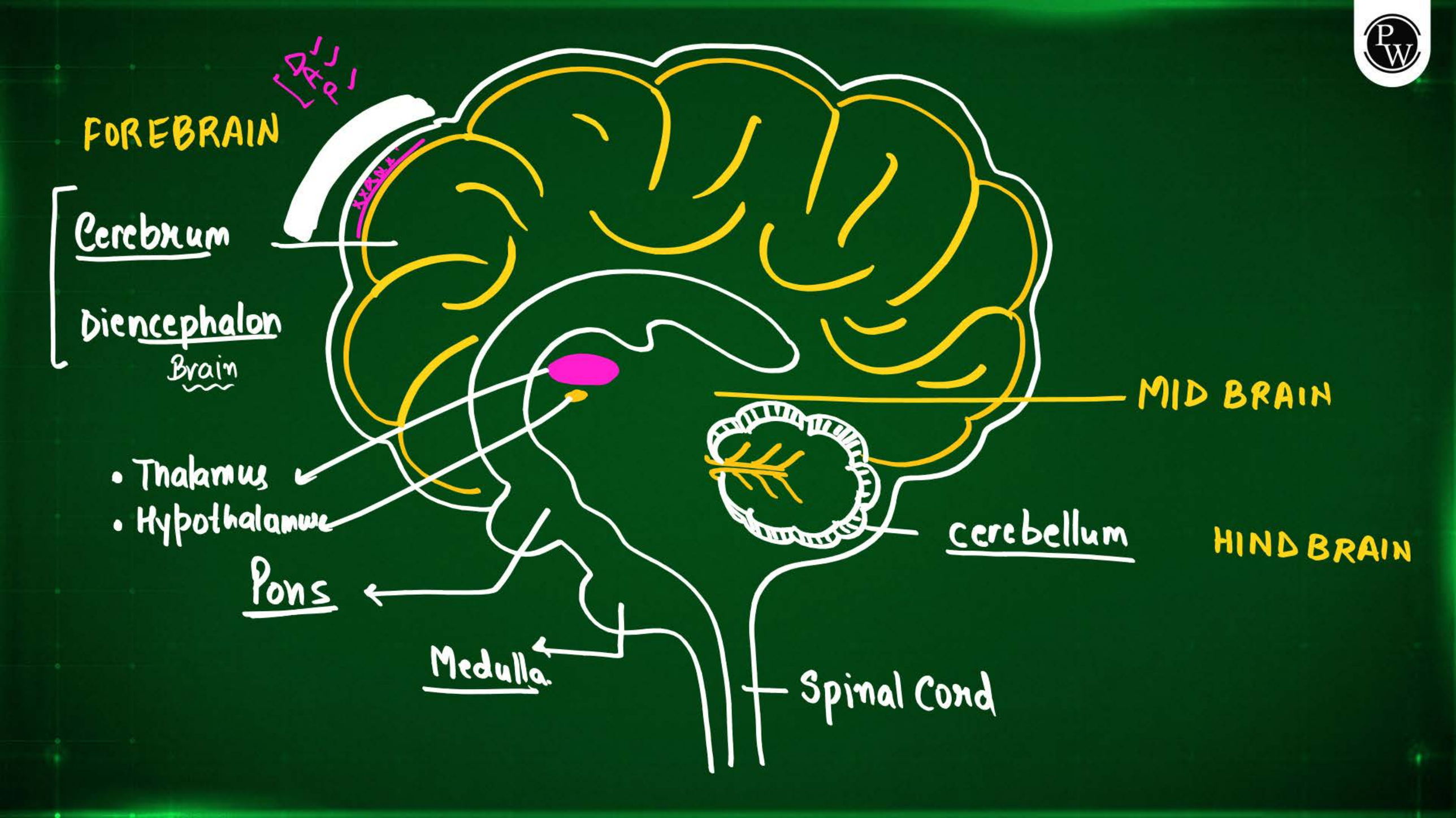
The space between the covering membranes is filled with a watery fluid –

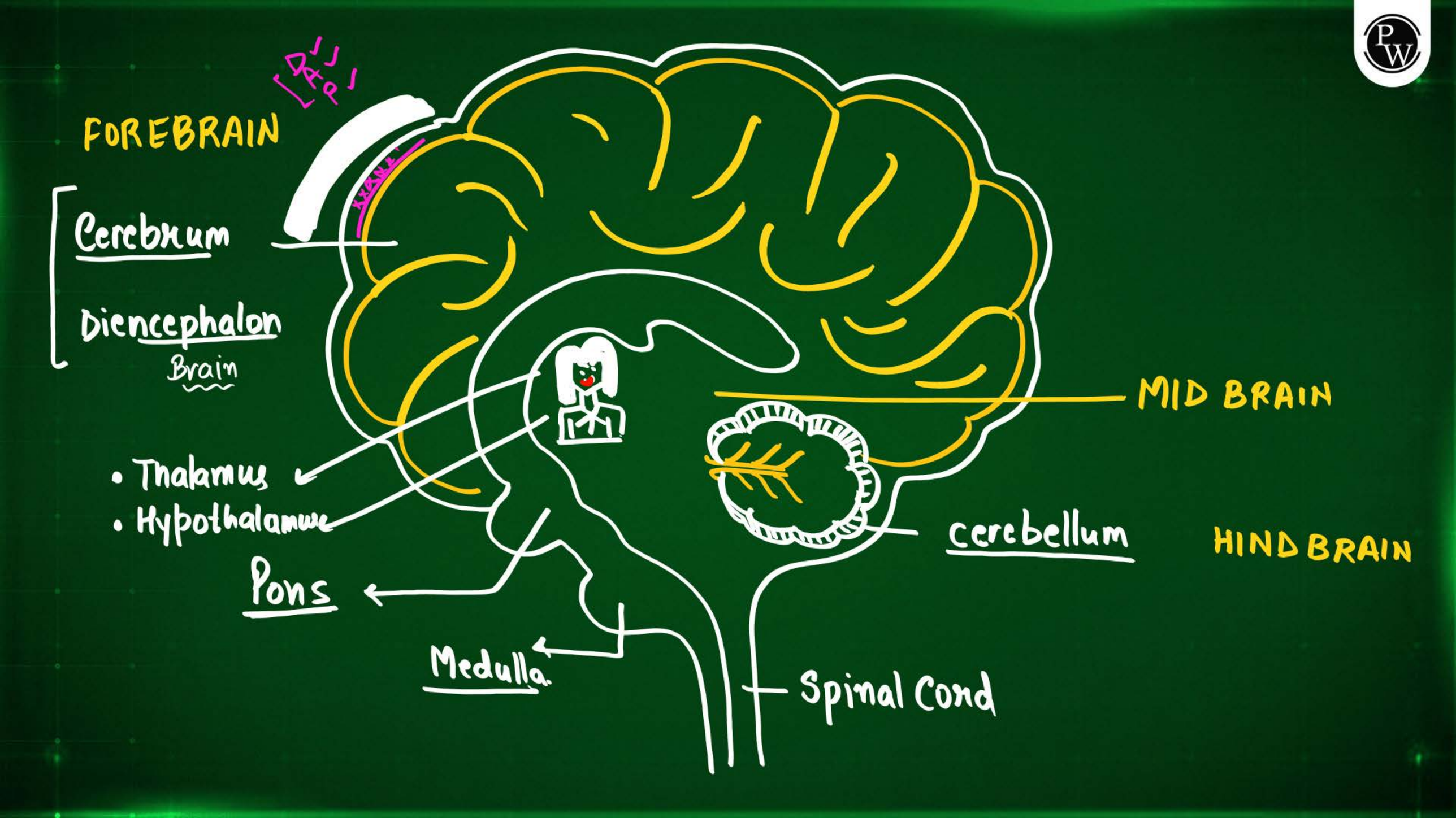
Cerebrospinal fluid (CSF) which acts like a cushion to protect the brain from shocks.

THE BRAIN

Three Primary Regions of the Brain

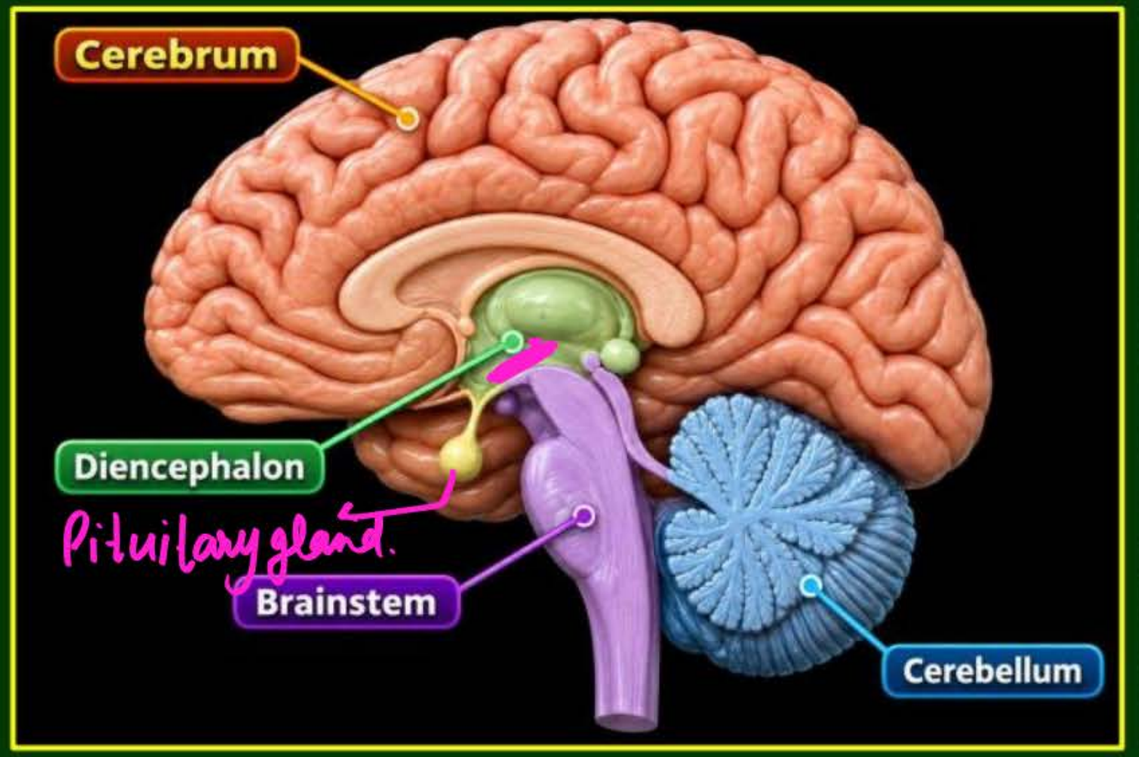
- A. Forebrain
  - 1. Cerebrum
  - 2. Diencephalon
    - Thalamus
    - Hypothalamus
- B. Midbrain
- C. Hindbrain
  - 1. Cerebellum
  - 2. Pons varoli
  - 3. Medulla oblongata


FOREBRAIN
Cerebrum
Diencephalon
Brain
• Thalamus
• Hypothalamus
Pons
Medulla
Spinal Cord
MID BRAIN
cerebellum
HIND BRAIN


FOREBRAIN
Cerebrum
Diencephalon
Brain
• Thalamus
• Hypothalamus
Pons
Medulla
MID BRAIN
cerebellum
HIND BRAIN
Spinal Cord

FOREBRAIN

Cerebrum (cerebral hemispheres) (seat of intelligence, memory, consciousness, will power, voluntary actions).



Diencephalon

- (a) Thalamus (relays pain and pressure impulses to cerebrum).
- (b) Hypothalamus (controls the body temperature and pituitary).

[37°C / 98.7°F]

control → thirst & hunger

Functions of brain

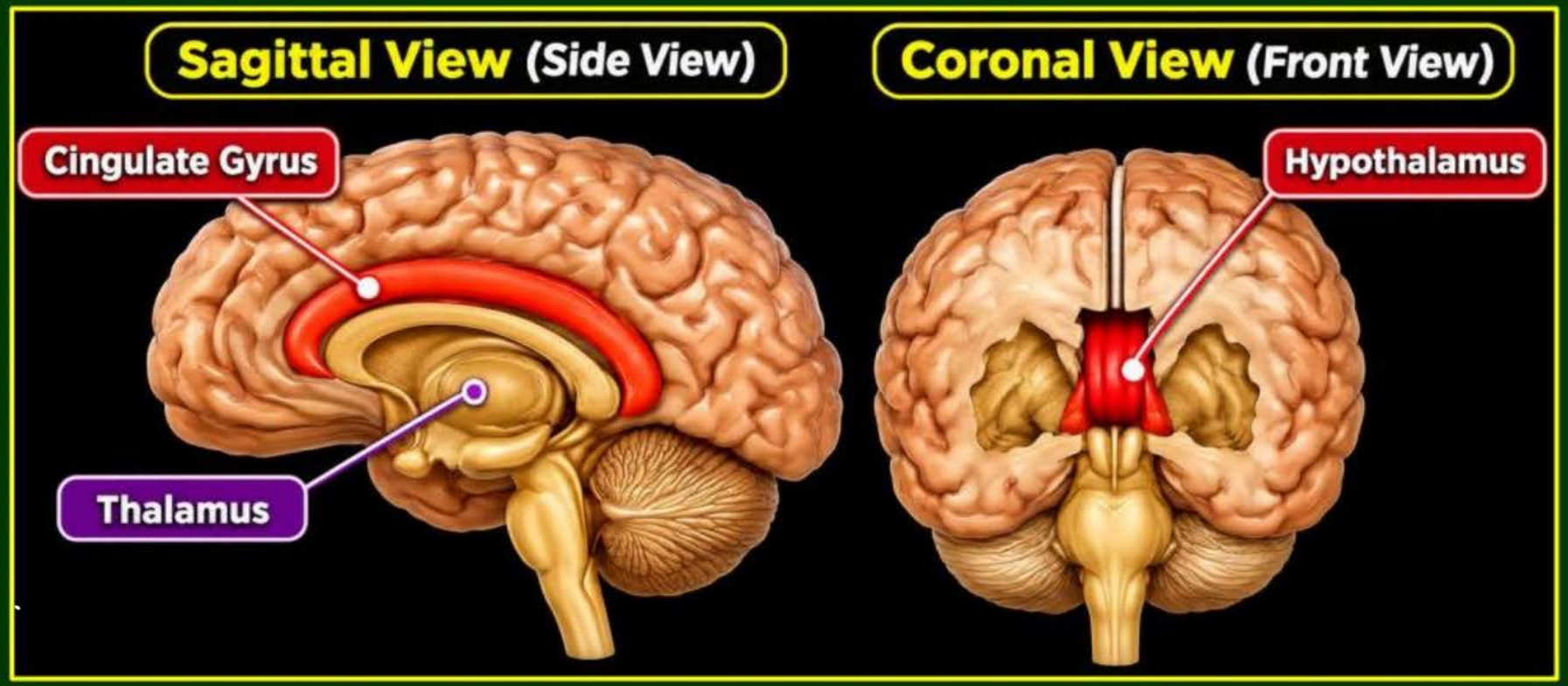
Information processing center:
Command and control center:

Involuntary action-
Heart, kidneys , lungs
Controls- thirst, hunger, emotions, anger, excitement , balance
Circulation rhythm- 24hrs cycle
Hormonal secretion

Voluntary action-

PARTS OF THE BRAIN

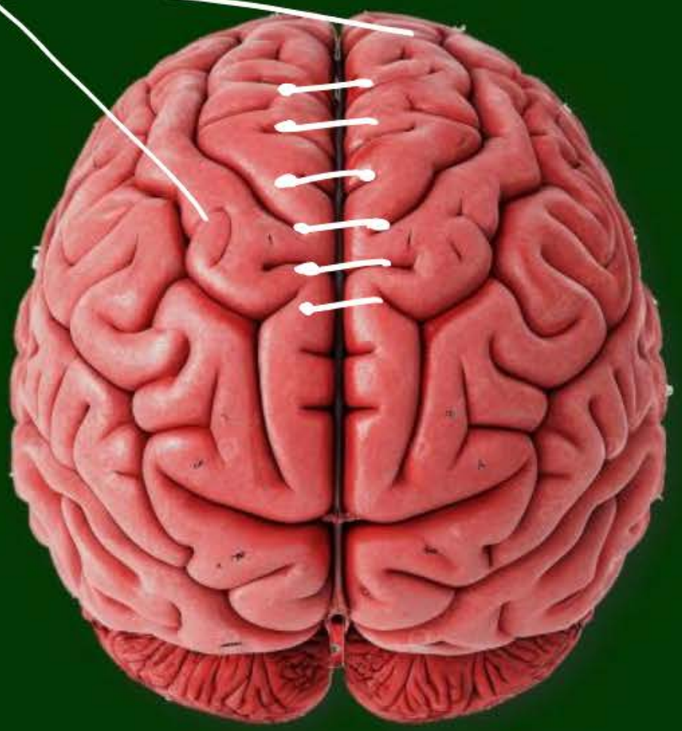
- Corpus callosum (hard body) is a sheet of fibres connecting the two cerebral hemispheres.
- Its function is to transfer information from one hemisphere to the other.



Functions of brain

2 Cerebral Hemisphere
→ left
→ Right

[CORPUS CALLOSUM] *





Read



LOBES IN CEREBRUM

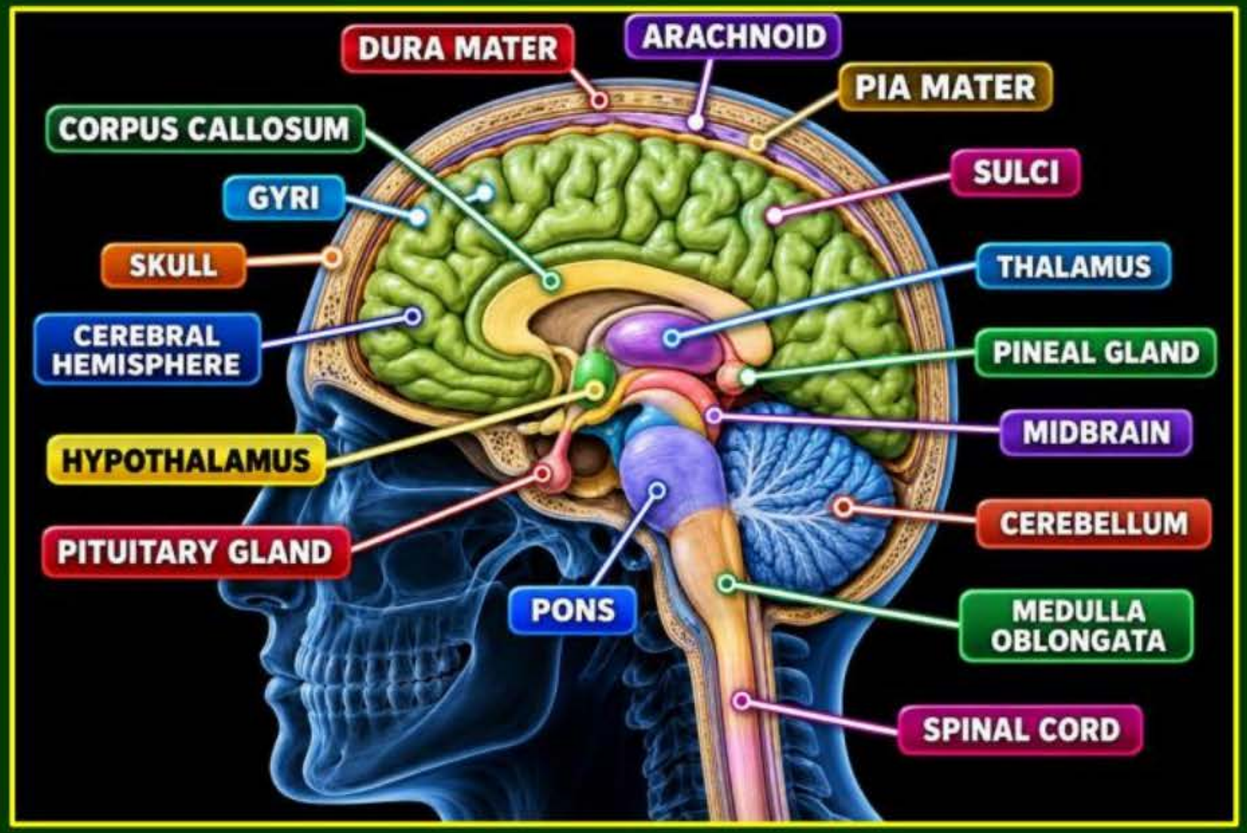
- Parietal
- Frontal
- Temporal
- Occipital
- Olfactory

PARTS OF THE BRAIN

Cerebrum (cerebrum : brain)

- Largest portion of the brain.
- Divided into two (right and left) halves called cerebral hemispheres.
- Their outer surface is highly convoluted with ridges and grooves.





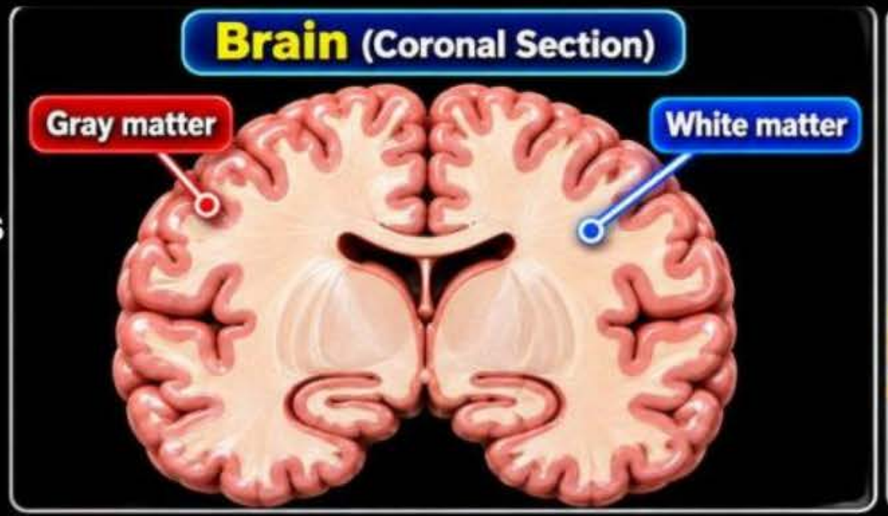
PARTS OF THE BRAIN

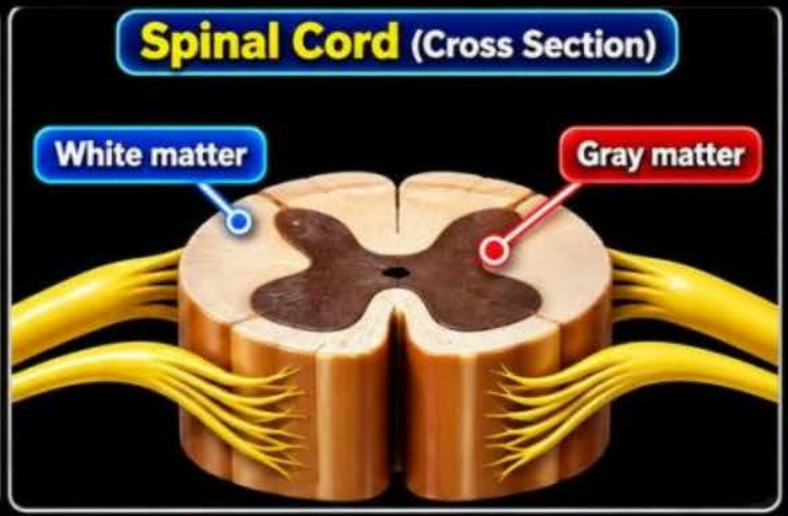
GREY MATTER

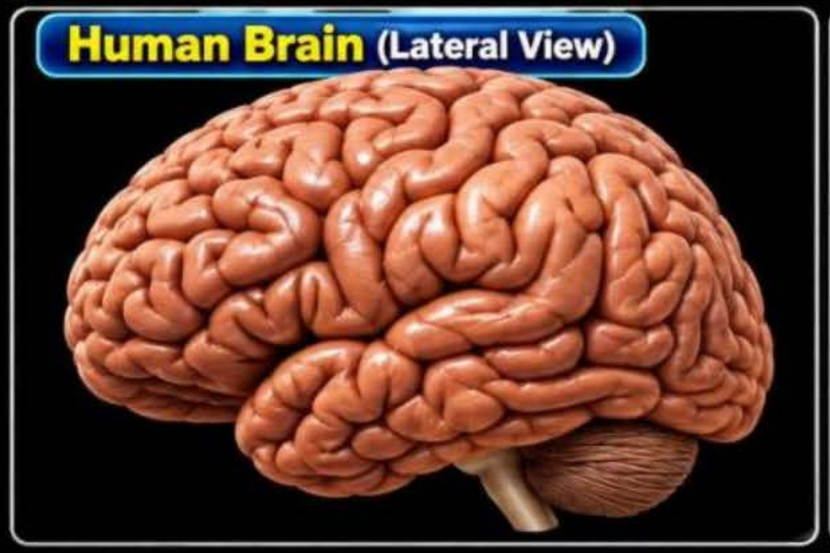
- Found in outer cortex region cerebrum
- Contains cell bodies of the neurons
- Appears greyish in colour,
- Folded to form the convolutions.
- The folds are called gyri and the grooves are called sulci (creases).

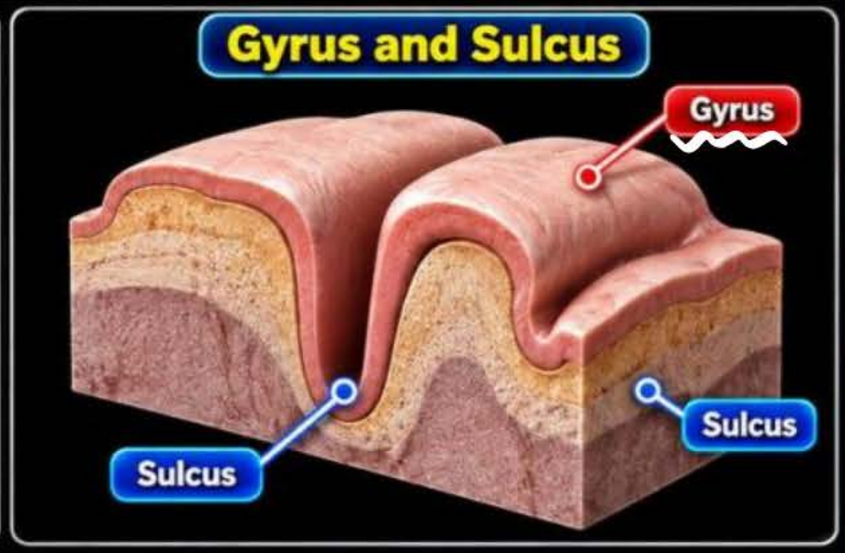
WHITE MATTER

- Inner portion of the cerebrum
- Contains mainly the axons (nerve fibres) of the neurons.









MIDBRAIN

- A small tubular part (reflexes involving eyes and ears).
- The midbrain controls reflex movements of the head, neck and trunk in response to visual and auditory stimuli.
- It also controls the reflex movements of the eye muscles, changes in pupil size and shape of the eye lens.

HINDBRAIN

(i)Cerebellum located just at the base and under the large cerebrum.

- (coordinates muscular activity, balance of the body),
- But has convolutions
- Gray matter in outer cortex and white matter inside takes impulse from cerebrum.



(ii) Pons located in the center of the brain and in front of the cerebellum

- (carries impulses from one hemisphere of the cerebellum to the other hemisphere and respiration).

(iii)Medulla oblongata lowest portion of the brain located at the base of the skull (controls involunatary activities of internal organs,
Regulation of respiration, regulation of cardiovascular activities (heart-beat) breathing, vomit center, gastric secretion etc.

CEREBELLUM (LITTLE BRAIN)



Fig. : Brain in median (sagittal) section

QUESTIONS

The inner covering of human brain is:

A Coracoid

B Piamater ✓

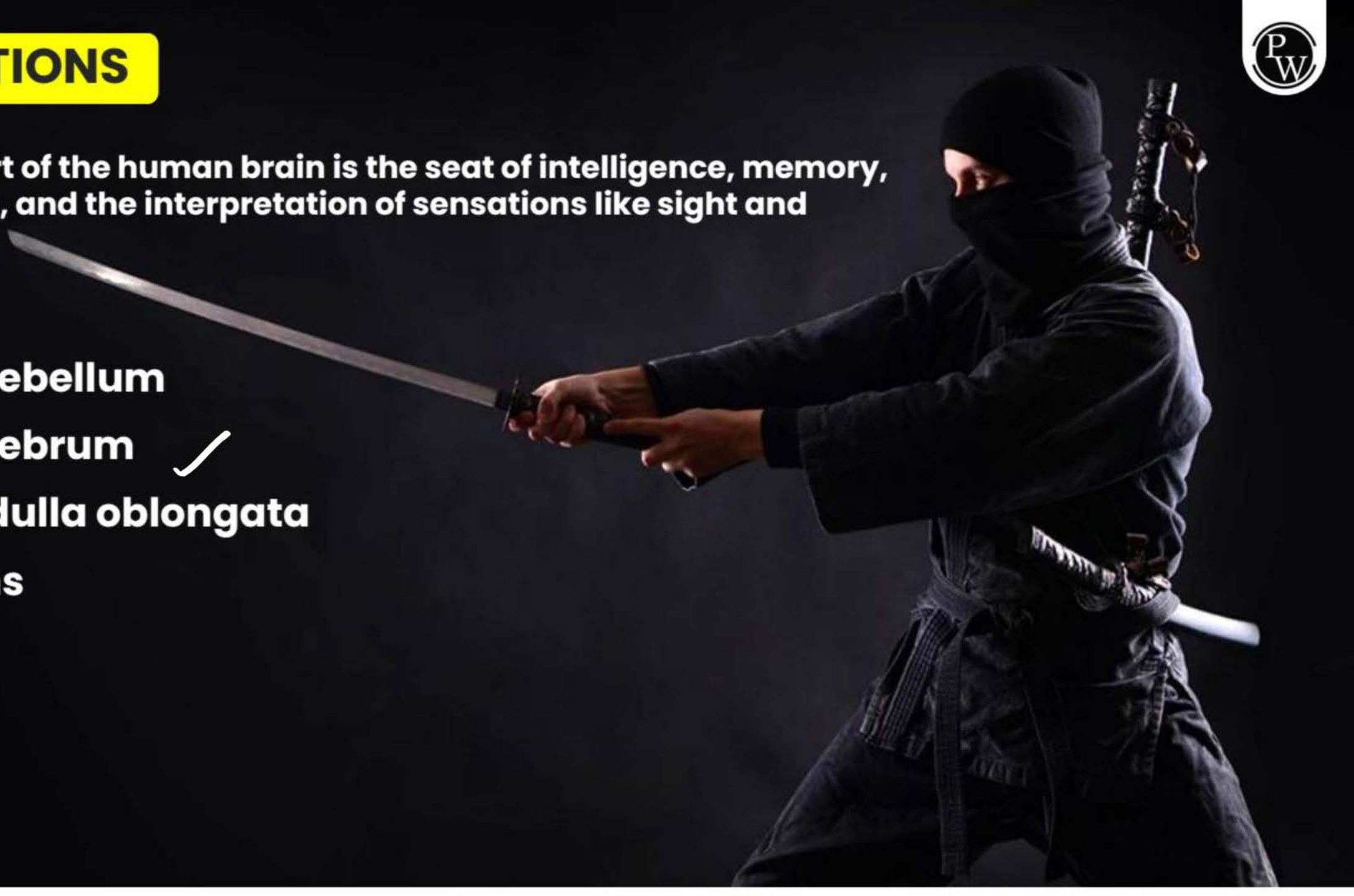
C Duramater

D Arachnoid

QUESTIONS

Which part of the human brain is the seat of intelligence, memory, reasoning, and the interpretation of sensations like sight and hearing?

- (a) Cerebellum
- (b) Cerebrum
- (c) Medulla oblongata
- (d) Pons



QUESTIONS

The two cerebral hemispheres are interconnected by a thick sheet of nerve fibers called the:

(A)Corpus luteum

(B)Corpus callosum

(C) Pons varolii

(D)Meninges



QUESTIONS

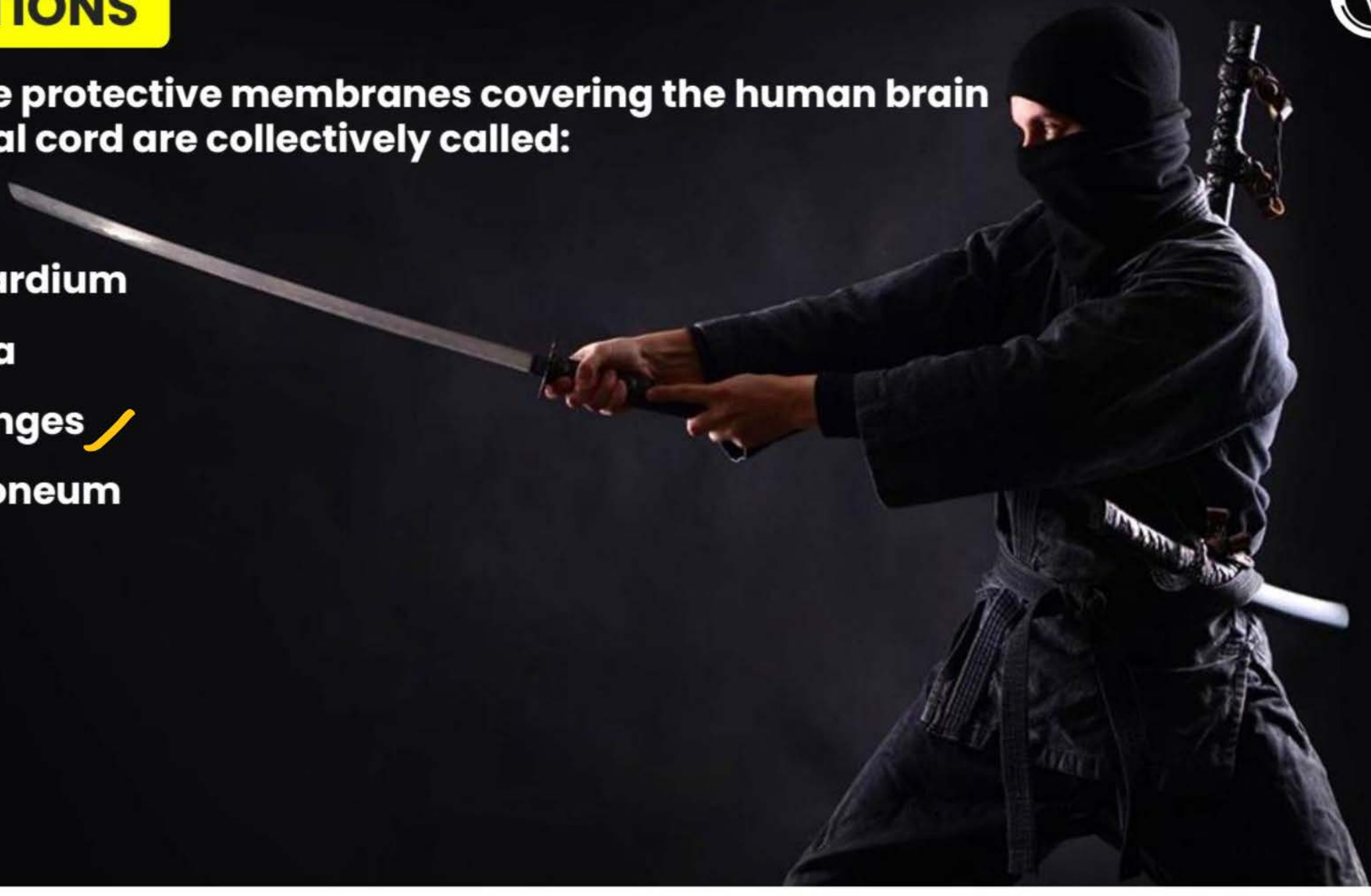
The three protective membranes covering the human brain and spinal cord are collectively called:

(A) Pericardium

(B) Pleura

(C) Meninges

(D) Peritoneum



QUESTIONS

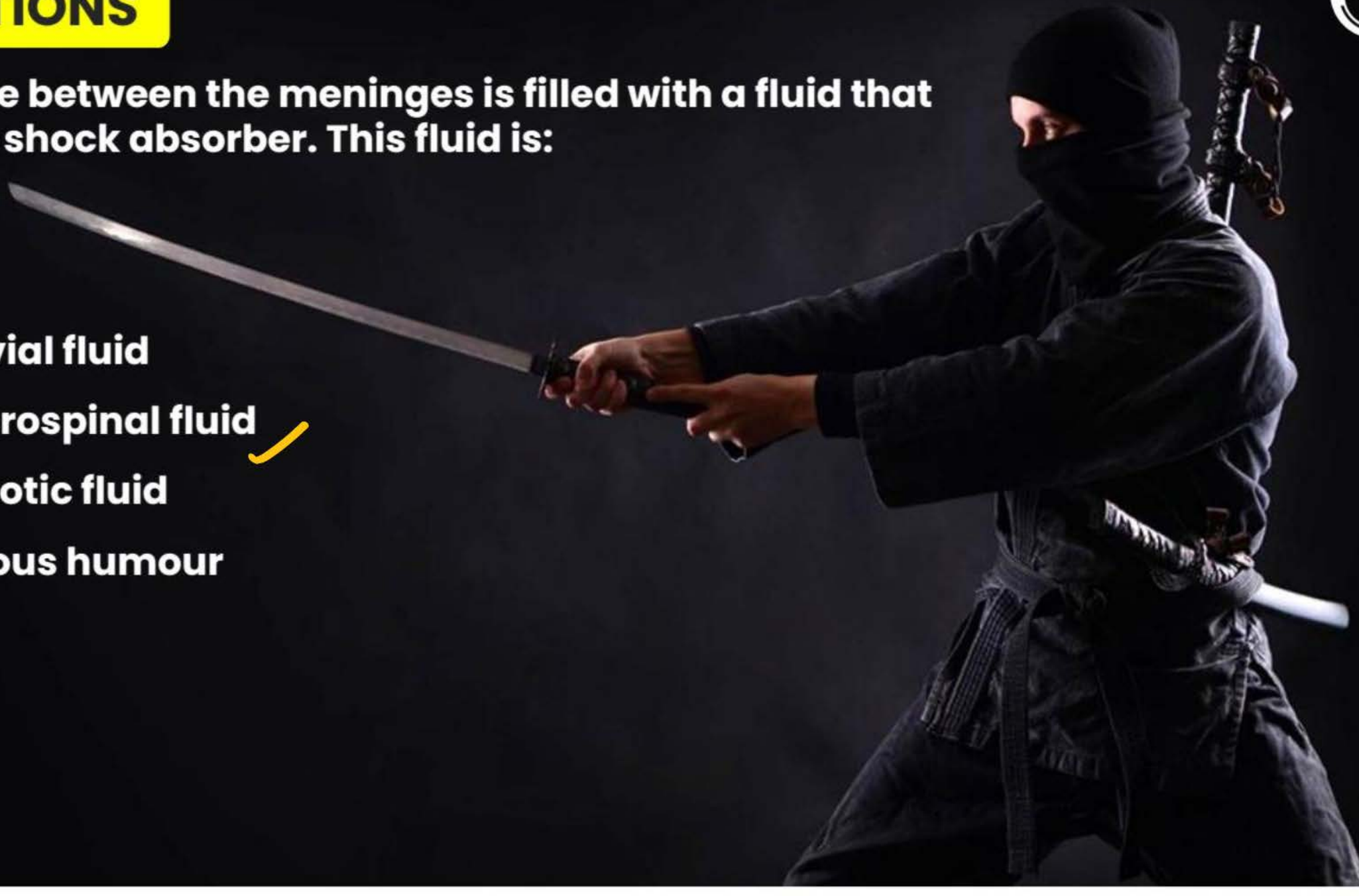
The space between the meninges is filled with a fluid that acts as a shock absorber. This fluid is:

(A)Synovial fluid

(B)Cerebrospinal fluid

(C) Amniotic fluid

(D) Vitreous humour



Thank You

